HACK APERTUS LIEBEFELD · RESEARCH & EXPERTISE NAVIGATOR CHALLENGE

BearMiner

Dig Bern gold from life-science partners.

A Research & Expertise Navigator that turns scattered university web pages into source-backed experts, units and partners — powered by Apertus.

BUILT FOR Bern Economic Development Agency

TEMPTEAM · OCTOBER 2026



WHY BERN

Switzerland's federal city, with a medtech engine behind it.

The canton's ambition

The Canton of Bern aims to be a leading international medical location by 2030, pooling the hospital, the university and industry.

Bern Economic Development Agency

The agency helps companies settle and grow in the canton: location advice, financing support, its partner network and events.

The gap BearMiner fills

The agency needs to know, fast, which companies fit Bern and which Bern experts to put them in touch with.

Inselspital

Switzerland's largest university hospital

University of Bern

Ranked among the world's top 100

sitem-insel

National competence centre for translational medicine

ARTORG Center

Biomedical engineering, built with Inselspital

Diabetes Center Berne

Diabetes technology, research and projects

Industry

Ypsomed, CSL Behring and the Bern medtech network

THE PROBLEM

To find an expert today, you have to know **where** to look before you know **what** to look for.

"glucose sensors" | continuous glucose monitoring | CGM | biosensing | metabolic monitoring | wearable sensing

3

source ecosystems: UniBE subdomains, ARTORG, Diabetes Center Berne

5+

page types per institute: faculty, group, person, project, service

0

places a company can ask one question and get traceable answers

OUR SOLUTION · HOW IT WORKS

Two sides of the same search, meeting on evidence.

RESEARCH SIDE

What Bern can do

- RULES Crawl UniBE, ARTORG and Diabetes Center Berne politely
- RULES Keep findings, projects and services, each with a quote
- RULES Up to 500 ranked highlights with EN/DE medical keywords

↔

MATCH

COMPANY SIDE

Who Bern should call

- APERTUS Reads any pitch, any language, e.g. Chinese
- APERTUS Builds a medical profile, every item quoted
- RULES Scores 0–100 with coins and P1–P4 priority

COMPANY SIDE · OUR LOGIC

From any pitch to a Bern match score.

1	**Read** Pitch PDF, website or text, any company	RULES
2	**Translate** Any language to English, e.g. Chinese sources	APERTUS
3	**Medical profile** Indications and needs, each quoted from the source	APERTUS
4	**Match** Themes vs Bern capabilities and crawled records	RULES
5	**Score** Six capped parts, 0–100, checkable by hand	RULES
6	**Hand off** P1–P4 queue with the Bern units to call first	RULES

MATCH SCORE = SIX PARTS, MAX 100



25+
a match

50+
gold match · P2

70+
very strong · P1

Not life science? Score 0, never a made-up partial match.

WHAT A RESULT LOOKS LIKE

Every answer carries its own proof.

DIABETES CENTER BERNE · PROJECT

The Moonwalk Project

glucose · hypoglycemia

> “After the end of the study, algorithms will be developed to build a warning system for nocturnal hypoglycemia…”

dcberne.com/en/project/the-moonwalk-project/

Real record from our live crawl test: keywords, verbatim quote, public source.

SAMPLE COMPANY · WEARABLE CGM, ZURICH

85.5



Very strong · P1 — prioritise outreach

Capability overlap	16.3/25
Partnership readiness	17.6/20
Medical focus	13.9/15
Domain fit	14.2/15
Swiss proximity	13.5/15
Signal depth	10/10

Why not higher: “Swiss company but no Canton of Bern site yet.”

PURPOSEFUL AI · TRACEABLE BY DESIGN

Apertus where language is hard. Rules where trust matters.

Apertus does

Bridges vocabulary — “glucose sensors” finds CGM, biosensing, metabolic monitoring

Reads companies — any-language pitch to a medical profile

Explains matches — a short, sourced reason per result

Code does

Decides the score — same input, same number, offline

Checks every claim against the source text

Collects and ranks research without a model or GPU

Quote or drop

No source quote (≥ 85% match), no item

Closed choice

Only candidates it was given, with URLs

Honest gaps

“No reliable match” instead of a guess

Verified location

Bern points never from model memory

TECHNICAL RIGOUR

The score separates real fits from noise.

SAMPLE COMPANY	COINS	SCORE
Glucose-sensor start-up, Burgdorf (BE)	💰💰💰	87.0
Wearable CGM company, Zurich	💰💰💰	85.5
CGM maker, Chinese-language source	💰💰💰	72.0
Surgical navigation, Munich	💰💰💰	71.8
Biopharma (oncology), Chinese-language source	💰	42.1
Fintech — out of scope	—	0.0

Our scorer on the repo samples: no LLM profile, no crawled Bern records loaded; Chinese text translated first. Backed by 30 labelled benchmark examples and tests for every evidence gate.

CURRENT DATA INVENTORY

42 source folders: UniBE departments and institutes, incl. ARTORG and Diabetes Center Berne

15,299 unique URLs in the crawl manifests (15,534 entries, 14,659 with HTTP 200)

3,775 HTML snapshots stored locally, about 162 MB

5,675 PDF links listed; no PDFs downloaded

SOVEREIGN · AFFORDABLE · EXTENSIBLE

Runs on Bern's terms, air-gapped if needed.

Sovereign

Apertus v1.5 on an on-premise or Swiss-cloud endpoint also handles translation. Sovereign deployments switch off the Google Translate fallback (EXTERNAL_MT=0).

Low cost

Crawling and scoring need no GPU. LLM answers are cached, so re-runs are instant and identical.

Extensible

A new institution is one config entry — then re-run crawl and enrich. No pipeline rewrite.

```
# src/sources.json — add a source
{ "id": "newinst", "name": "New Institute", "url": "https://…", "institution": "…" }
```

FROM PROTOTYPE TO PILOT

Give us one quarter and one real company inbox.



A TOOL FOR Bern Economic Development Agency

WEEKS 1–4

Complete crawl

Full run over all three ecosystems on a Swiss-hosted Apertus endpoint.

WEEKS 5–8

Expert review

Agency staff review highlights; feedback tunes rules and synonyms.

WEEKS 9–12

Live outreach

Score incoming company requests and hand off P1 leads to Bern units.